E14e - Phase Shift in AC Circuits

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May 20, 2025

Abstract

This report examined the evolution of the phase angle over a changing frequency in order to extract key circuit parameters for RLC, RL and RC AC circuits from theoretical models. All circuits expressed a coefficient of determined, $r^2 > 0.997$ against models and displayed the key phase characteristics for there respective circuit. From the RC circuit parameters $R = 176.136 \pm 0.777\ \Omega$ and $C = 1.013 \pm 0.005\ \mu F$ possessed a relative error of $\approx 1.71\%$ and $\approx 0.70\%$, with experimental results $R_{\text{exp}} = 179.2\ \Omega$ and $C_{\text{exp}} = 1.006\ \mu F$ respectively, lying within acceptable margins and confirming the validity of the fit. The inductance $L = 4.159 \pm 0.047\ mH$ was determined from the RL fitting with a relative error of $\approx 3.73\%$ from the experimental $L_{\text{exp}} = 4.32\ mH$ value measured. The RLC analysis yielded a fitted resonant frequency $f_0 = 2414.233\ Hz$, with $r^2 = 0.999723$ translating to a fitted Inductance $L = 4.32\ mH$ and total resistance $R_{\text{total}} = 182.255 \pm 0.001\ \Omega$ and thus $R_{\text{sp}} = 6.119\ \Omega$, possessing a relative error of $\approx 5.5\%$ from experimentally measured $R_{\text{sp,exp}} = 5.8\ \Omega$. The discrepancy between determined parameters and experimental values are likely a result of measurement uncertainty or systematic noise that was not considered.

1 Introduction

Alternating current (AC) circuits exhibit a range of behaviors not present in direct current (DC) systems, one of the most important being the phase shift between voltage and current. This phase shift, defined as the angular difference in time between corresponding points of the voltage and current waveforms, plays a crucial role in understanding the energy flow, impedance characteristics, and frequency response of AC circuits. It is fundamental to the analysis of signal propagation, power distribution, resonance, and filtering in electrical and electronic systems.

In this experiment, we investigate the phase behavior of three elementary AC circuit configurations: the **RC (resistor-capacitor)**, **RL (resistor-inductor)**, and **RLC (resistor-inductor-capacitor)** series circuits. Each configuration exhibits a distinct relationship between frequency and phase shift due to the differing reactive properties of capacitors and inductors.

The primary objective of this experiment is to measure the phase shift ϕ between the applied voltage and the current (inferred from the voltage across a series resistor) as a function of frequency for each circuit type. By analyzing the phase response, we aim to extract key circuit parameters (such as resistance, capacitance, and inductance) and compare these with independently measured values. The experimental data is compared with theoretical predictions based on phasor analysis and impedance models.

Through this study, we gain both theoretical and practical insight into the frequency-dependent dynamics of AC circuits, enhancing our understanding of complex impedance and the critical role of phase in real-world electrical systems.

2 Theoretical Analysis

Phase shift is the horizontal, (typically time), displacement between two waves, Fig.1. The phase shift is typically measurement as a radian or a degree.

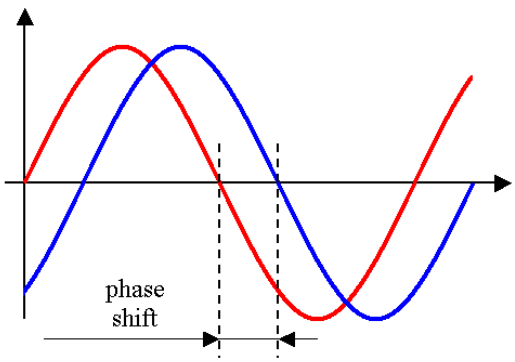

Figure 1: Phase Shift between two waves

Source: [1]

For the RLC, RC and RL circuits used in this experiment, Fig.2. the ideal phase shift can be determined by Eq.1, which itself is a result of a vertical projection of a phasor circuit-analysis[2].

$$\phi_{\text{RLC}} = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R_{\text{tot}}}\right)$$

$$\phi_{\text{RL}} = \arctan\left(\frac{\omega L}{R_{\text{tot}}}\right) \tag{1}$$

$$\phi_{\text{RC}} = \arctan\left(\frac{-\frac{1}{\omega C}}{R_{\text{tot}}}\right)$$

Where:

ϕ :	Phase shift (rad)
ω :	Angular Frequency $(\frac{\text{rad}}{s})$: $\omega = 2\pi f$
f :	Frequency (Hz)
L :	Inductance (H)
C :	Capacitance (C)
R_{tot} :	total resistance (Ω)

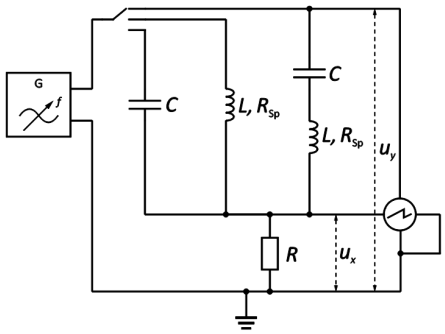

Fig. 1 Circuit for task 1
G: Arbitrary waveform generator
C: capacitor with capacitance *C*
L, R_{sp}: coil with inductance *L* and series resistance R_{sp}
R: resistor with resistance *R*
U_x, U_y: oscilloscope channels 1 and 2

Figure 2: Interchangeable RLC, RL and RC circuit diagram
Source: [3]

For the case of the RLC circuit, from ϕ_{RLC},Eq.(1). it is expected that the circuit will have a resonant frequency that is dependent on the impedance of the capacitor,$Z_C = \frac{1}{\omega C}$, and inductor ωL, in particular the point at which they are equal leading to the relation:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \tag{2}$$

However it is expected that the RL and RC circuit do not possess a resonant frequency.

3 Experimental Analysis

3.1 Experimental Setup

The experimental setup was designed to measure the phase shift ϕ between current and voltage as a function of frequency in three types of AC series circuits: an RLC circuit, an RC circuit, and an RL circuit. In all three configurations, the current was not directly measured. Instead, it was inferred by measuring the voltage drop across a known resistor placed in series with the other components, (Ohm's Law). This voltage is in phase with the current and therefore serves as a reliable proxy.

The setup consisted of an arbitrary waveform generator, which provided a sinusoidal voltage signal of adjustable frequency and amplitude. This signal was applied across the complete series circuit under investigation. A resistor with known resistance R was connected in series with either a capacitor, an inductor, or both, depending on the circuit being studied. The inductor used in the RL and RLC configurations included an internal series resistance, denoted as R_{sp}. A two-channel digital oscilloscope was employed to monitor the voltage signals at two points in the circuit: the voltage across the resistor (U_x), and the total voltage across the entire circuit (U_y). These signals were connected to channels 1 and 2 of the oscilloscope, respectively. This setup is shown is Fig. 2

For the RC series circuit, the voltage source was connected in series with the resistor and capacitor. For the RL series circuit, the setup included the resistor and the coil with its internal resistance. Finally, the RLC series circuit consisted of all three components—resistor, inductor (with R_{sp}), and capacitor, connected in series. The total voltage across the circuit represented the input voltage from the waveform generator.

To determine the phase shift, the time difference Δt between the zero crossings (or peaks) of U_x and U_y was measured using the cursor functionality of the digital oscilloscope. The phase shift was then calculated using the equation:

$$\phi = 2\pi f \Delta t \tag{3}$$

In the case of the RLC circuit, the resonance frequency was first identified using Lissajous figures. This method involved switching the oscilloscope to X-Y mode, where the horizontal axis displayed U_x and the vertical axis displayed U_y. At resonance, the phase shift between current and voltage becomes zero, resulting in a straight diagonal line on the oscilloscope screen. This provided a clear and efficient way to identify the resonance point before conducting further frequency-dependent phase shift measurements. The frequency range for all measurements was chosen to be neatly spaced to ensure high resolution near resonance or other regions where the phase changed rapidly.

3.2 Data Analysis

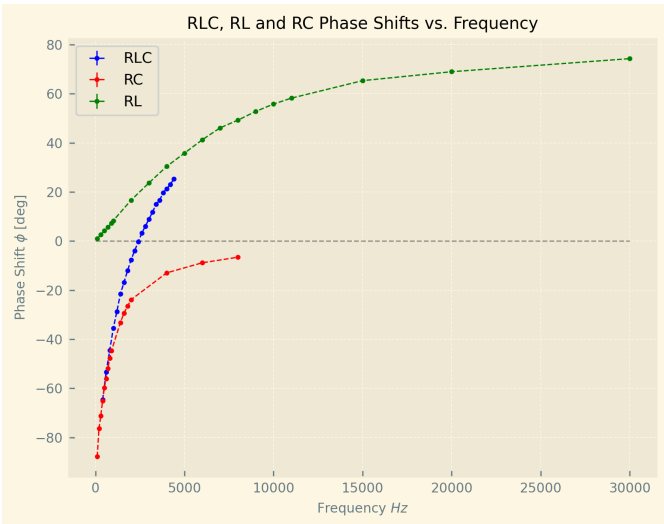

Figure 3: Phase Shift of each RLC, RL and RC circuits

The phase shift over a range of frequency measurements for each of the RLC, RL and RC circuits is displayed in Fig.3. and, despite the limited range of the RLC, and RC measurements show typical behavior for each of the respective circuits.
Fig.4 shows the phase shift versus frequency for the RLC, RC and RL circuits on a logarithmic scale. This proves useful in observing the trend of these circuits under changing frequencies. The range of frequency was extended for the fitted curves in order to allow a detailed view of both low- and high-frequency behavior. The dotted reference lines at $-90°$, $0°$, and $+90°$ highlight

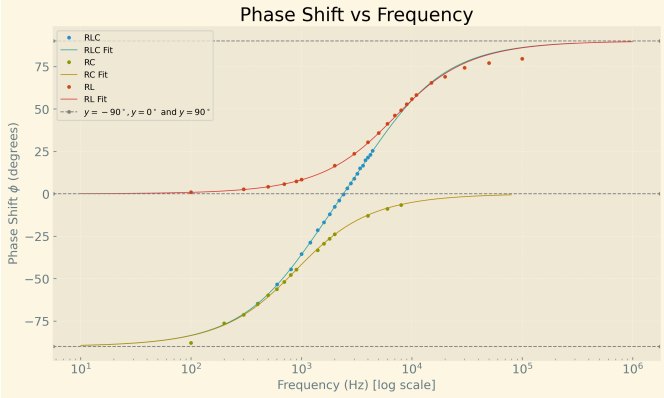

Figure 4: Phase Shift of circuit on a logarithmic scale

the asymptotic limits predicted by theory.

For the RLC circuit, the phase shift is governed by the expression $\phi = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$. At low frequencies, the $\frac{1}{\omega C}$ term dominates, resulting in a large negative argument of the arctangent function. This causes the phase shift to approach $-90°$, indicating a behavior similar to that of a pure capacitor. At high frequencies, the ωL term becomes dominant, leading the phase to approach $+90°$, as expected for an inductor-dominated regime. In between, at the resonance frequency where $\omega L = 1/(\omega C)$, the net reactance vanishes and the phase shift passes through $0°$. The experimental data for the RLC circuit follows this transition smoothly, and the fitted curve captures the characteristic S-shaped response. The phase points appear more spread out in the low-frequency (capacitive) region and more compressed in the high-frequency (inductive) region. This asymmetry is not due to how the data was collected—since the frequency values were chosen symmetrically around the resonance frequency f_0 and incremented with equal steps—but rather arises from the nonlinear nature of the arctangent function and the different sensitivities of the phase response in the two regimes. In the $\frac{1}{\omega C}$-dominated region, small changes in frequency cause large changes in phase, leading to wider spacing. In contrast, in the ωL-dominated region, phase changes more slowly, resulting in closer clustering of points.

The RC circuit exhibits a monotonic increase in phase shift with frequency, described by the function $\phi = \arctan\left(-\frac{1}{\omega RC}\right)$. At low frequencies, the capacitive impedance is large, making the magnitude of the negative argument of the arctangent large as well, which drives the phase shift toward $-90°$. As frequency increases, the impedance of the capacitor decreases, reducing the phase difference between voltage and current. Consequently, the phase shift gradually rises toward $0°$, representing a transition to a more resistive behavior.

For the RL circuit, the phase shift is defined by the relation $\phi = \arctan\left(\frac{\omega L}{R_tot}\right)$. At low frequencies, the inductive reactance is small, and the phase shift remains near $0°$, indicative of a resistive response. As the frequency increases, the reactance increases, and the phase shift rises accordingly, asymptotically approaching $+90°$ in the high-frequency limit.

3.2.1 RC Analysis

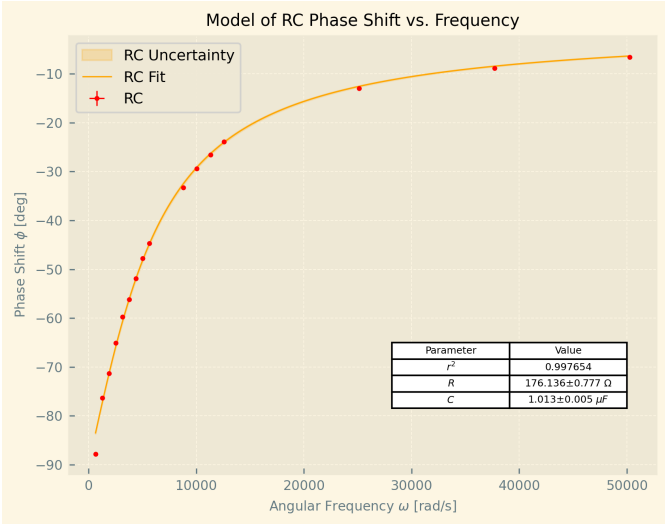

Figure 5: Phase Shift Modeling of RC circuit

In Fig. 5, the RC circuit exhibits a characteristic negative phase shift that approaches $0°$ in the high-frequency limit. The experimental phase shift data was fitted to the theoretical expression for the phase angle of an RC series circuit given by the equation in the set of Eq.(1). From the fit, the resistance was determined to be $R = 176.136 \pm 0.777\ \Omega$ and the capacitance $C = 1.013 \pm 0.005\ \mu$F. The coefficient of determination, $r^2 = 0.997654$, indicates an excellent agreement between the measured data and the theoretical model. The experimentally measured series resistance was $R_{\text{exp}} = 179.2\ \Omega$, and the measured capacitance was $C_{\text{exp}} = 1.006\ \mu$F. Although both of these values lie outside uncertainties of their respective parameter values, the closeness between the experimental and fit values is apparent. The relative error between the fitted and experimental resistance is approximately $\approx 1.71\%$, while for the capacitance it is $\approx 0.70\%$.

Both deviations lie within acceptable margins and confirm the validity of the fit. The small uncertainty and high r^2 underscore the consistency of the RC circuit measurements with the theoretical model.

3.2.2 RL Analysis

The RL circuit displayed a steadily increasing phase shift as the frequency increased, asymptotically approaching $+90°$. The phase shift data was fitted according to the equation for RL circuit in the set of Eq. (1), yielding an inductance of $L = 4.159 \pm 0.047$ mH. The correlation coefficient of the fit was $r^2 = 0.998836$, again indicating a highly reliable model fit. The experimental value of the inductance was $L_{\text{exp}} = 4.32$ mH. The relative error in this case is $\approx 3.73\%$.

This discrepancy is within a reasonable experimental tolerance, and likely results from measurement uncertainty or systematic noise during high-frequency measurements that was not considered within the uncertainty calculated.

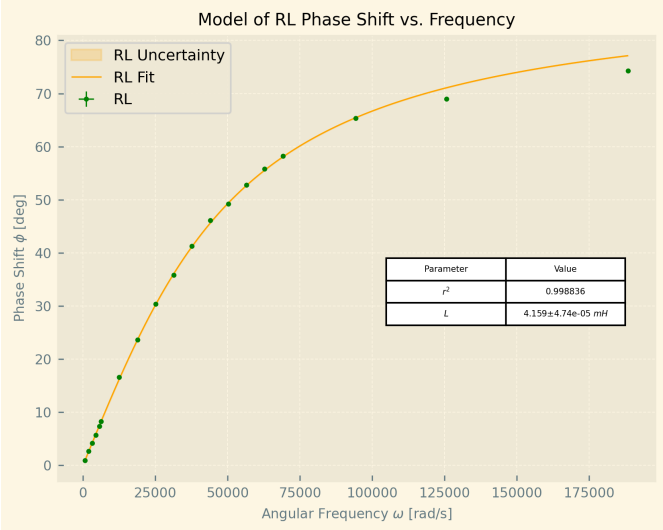

Figure 6: Phase Shift Modeling of RL circuit

The close match between the fitted and measured values, along with the tight uncertainty bounds and high r^2, supports the accuracy of the phase shift model for the RL circuit.

3.2.3 RLC Analysis

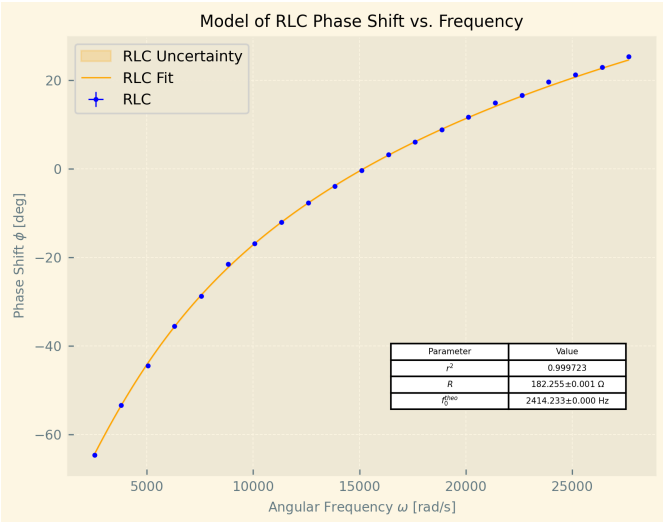

Figure 7: Phase Shift Modeling of RLC circuit

The RLC circuit phase shift curve in Fig.7 displays the typical transition from negative to positive phase angles, passing through zero at the resonance frequency. This frequency was determined using the Lissajous figure method to be $f_0 = 2402.9$Hz, corresponding to an angular frequency $\omega_0 = 2\pi \cdot 2402.9 \approx 15091.6$rad/s. From this resonance condition, the inductance can be calculated using:

$$L = \frac{1}{\omega_0^2 C} \tag{4}$$

With the experimental capacitance $C = 1.006\ \mu$F, the resulting inductance is $L = 4.35$mH. From Fig. 7, the fitted resonance frequency is $f_0 = 2414.233$ Hz, this translate to an inductance of $L = 4.32$ mH. the inductance L determined from the fitted resonance frequency is exactly the same value as the experimentally determined inductance L_{exp} and is in excellent agreement RL circuit fit value. Additionally, the phase shift data was fitted to the full RLC phase angle expression give in Eq. (1).

The fitted total resistance was $R_{\text{total}} = 182.255 \pm 0.001\ \Omega$ with excellent fit quality of $r^2 = 0.999723$. Since this total resistance includes both the series resistor and the coil's internal resistance R_{sp}, the internal resistance of the inductor can be determined by subtracting the fitted value of $R = 176.136\ \Omega$ obtained from the RC circuit analysis. This yields $R_{\text{sp}} = 6.119\ \Omega$, which compares well with the experimentally measured coil resistance of 5.8 Ω, with a relative error of $\approx 5.5\%$. This slightly larger deviation may result from thermal effects or minor contact resistance, but remains within acceptable bounds.

4 Discussion

The experimental investigation of phase shifts in RC, RL, and RLC circuits successfully demonstrated the expected theoretical behavior for each configuration, with the data and fitted parameters showing strong alignment with predictions. The primary aim was to quantify how the phase angle between the voltage and current evolves with frequency and to extract key circuit parameters through model fitting. Across all three circuit types, the high coefficients of determination ($r^2 > 0.997$ in all cases) provide strong evidence for the reliability of both the measurements and the underlying theoretical models.

For the **RC circuit**, the phase shift exhibited the anticipated monotonic progression from approximately $-90°$ at low frequencies toward $0°$ at higher frequencies. The resistance and capacitance values obtained from the fit differed from the experimental measurements by less than 2%, with relative errors of approximately 1.71% for R and 0.70% for C. These small discrepancies are well within acceptable experimental error and affirm the accuracy of the fit. Minor deviations could be attributed to systematic noise, parasitic capacitance, or limitations in manual cursor-based time shift measurements.

In the **RL circuit**, the phase shift increased with frequency as expected, showing a rise towards the $+90°$ asymptote in the high-frequency limit. The inductance extracted from the fit, $L = 4.159$ mH, matched closely with the experimentally determined value of 4.32 mH, with a relative error of about 3.73%. This small discrepancy is likely due to slight thermal effects on the inductor or limitations of the frequency resolution near the upper end of the measurement range. The very high $r^2 = 0.998836$ confirms the robustness of the fit despite these minor differences.

The **RLC circuit** produced the most comprehensive behavior, transitioning from capacitive to inductive dominance as the frequency passed through the resonance point. The resonance frequency obtained from the fitting curve $f_0 = 2414.233$ Hz was used to calculate an inductance of $L = 4.35$ mH, exactly the same as the experimentally determined inductance and in excellent agreement with the fit from the RL circuit. Additionally, the fitted total resistance of the RLC circuit, $R_{\text{total}} = 182.255, \Omega$, includes both the external resistor and the inductor's internal resistance. Subtracting the

known series resistance $R = 176.136\ \Omega$ yielded an internal coil resistance $R_{\text{sp}} = 6.119, \Omega$, which closely matched the measured value of 5.8 Ω with a relative error of 5.5%. The minor excess may be attributed to contact resistance at terminals or subtle resistive heating during operation.

Overall, the experiment strongly supports the theoretical models of phase behavior in AC circuits. The observed trends, the close numerical agreement of fit and experimental parameters, and the consistently high correlation coefficients all point to well-controlled measurements and correct model implementation. The discrepancies that do exist remain within a plausible margin for manual measurements and real-world circuit tolerances, highlighting both the precision and practical reliability of phase shift analysis in AC systems.

5 Conclusion

This experiment successfully explored the frequency-dependent phase shifts in RC, RL, and RLC series circuits, verifying the theoretical predictions for each configuration. By analyzing the time displacement between the voltage across the resistor and the total circuit voltage, the phase angle was determined over a wide frequency range and used to extract key circuit parameters.

The RC circuit exhibited a smooth transition from capacitive to resistive behavior, and the fitted resistance and capacitance closely matched the experimentally measured values with relative errors below 2%. The RL circuit demonstrated the expected inductive phase increase with frequency, and the fitted inductance showed excellent agreement with the measured value. The RLC circuit displayed a characteristic resonance behavior, with the resonance frequency identified via Lissajous figures. The inductance calculated from the resonance condition aligned well with both the measured and fitted values. Furthermore, the internal resistance of the inductor was accurately determined by subtracting the known series resistance from the total fitted resistance.

Across all circuits, the quality of the fits was high, as indicated by correlation coefficients exceeding 0.997. These results confirm the validity of the theoretical phase shift models and illustrate the effectiveness of phase measurements as a diagnostic tool for determining component values in AC circuits. The experiment not only reinforced theoretical understanding but also provided practical experience in circuit analysis, waveform interpretation, and quantitative modeling.

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